US Healthcare Payment Integrity

A Medicare Inpatient Payment Analysis · CMS Datasets 2017–2018

196,086 Hospital × DRG Records	**3,182** Unique Hospitals
7.4M Total Discharges	**51** US States / Territories

Analysed by: Apekshaa · Tools: Python · SQLite · Pandas · Matplotlib · March 2026

The Problem: US hospitals submit "sticker-price" bills to Medicare; Medicare pays a fraction. This project quantifies how lar

Notebook 01 · Data Understanding

3 raw CMS datasets explored
196,086 rows · 3,182 hospitals · 563 DRGs
Join keys identified & validated
Data grain confirmed: Hospital x DRG

Notebook 02 · Data Cleaning

Column names standardised
Billing_Gap & Charge_Ratio derived
MDC / MDC_Name columns preserved
Zero-payment rows removed

Notebook 03 · Database Setup

Cleaned CSVs loaded into SQLite
Joined view: vw_inpatient_analytics
All subsequent queries use this view
No server required — in-process DB

Notebook 04 · Python Analysis

Billing gap distributions
Charge ratio histogram (linear + log)
Complexity vs payment scatter
Ownership boxplot comparison

Notebook 05 · SQL Analysis

Window functions: DENSE_RANK
PERCENT_RANK · ROW_NUMBER
CTEs for readable queries
Geographic MDC 05 analysis

Notebook 06 · Case Study Answers

All 4 case-study tasks answered
3 independent findings from data
Each explained in plain English
With supporting chart and SQL

Tools used:

Python 3.11	pandas / numpy	SQLite + SQLAlchemy	matplotlib / seaborn	scipy	Jupyter
Core language	Data wrangling & stats	In-process database	Charts & heatmaps	Regression & correlation	6 notebooks end-to-end

Three CMS datasets and how they connect into the analytical view

T1 Inpatient Payments (196,086 rows)

Row grain: one hospital treating one DRG procedure type

- Rndrng_Prvdr_CCN — 6-digit hospital ID (join key to T3)
- DRG_Cd — procedure code (join key to T2)
- Tot_Dschrgs — number of patients treated
- Avg_Submtd_Cvrd_Chrg — what the hospital billed Medicare
- Avg_Mdcr_Pymt_Amt — what Medicare actually paid
- Billing_Gap = Billed - Paid (derived field)
- Charge_Ratio = Billed / Paid (derived field)

DRG_Cd = DRGV22

T2 DRG Details (518 procedure types)

One row per procedure type

- DRGV22 — join key to DRG_Cd in T1
- MDC / MDC_Name — body-system group (25 categories)
- Relative Weights — CMS complexity score (higher = more pay)
- Geometric Mean LOS — typical hospital stay (days)

CCN = Facility_ID

T3 Provider Info (5,426 hospitals)

One row per hospital

- Facility ID — join key to CCN in T1
- Hospital Type — Acute Care / Critical Access / Psychiatric
- Hospital Ownership — Non-profit / For-profit / Government

vw_inpatient_analytics (SQL VIEW)

T1 LEFT JOIN T3 on CCN · T1 LEFT JOIN T2 on DRG_Cd · 196,086 rows · All analysis queries use this view

Key columns in each table and what they mean in the real world

Each row in T1 is one hospital treating one type of procedure for some number of patients. T2 and T3 are lookup tables that add procedure complexity and hospital profile context.

T1 Inpatient Payments — the money table

Rndrng_Prvdr_CCN	Hospital's 6-digit Medicare certification number. Join key to T3.
DRG_Cd	Procedure code — which type of treatment. Join key to T2.
Tot_Dschrgs	Number of patients for this hospital + procedure combination.
Avg_Submtd_Cvrd_Chrg	"Sticker price" — avg amount the hospital billed Medicare.
Avg_Mdcr_Pymt_Amt	Avg amount Medicare actually paid (after all adjustments).
Billing_Gap (derived)	Billed minus Paid. Dollar gap per patient.
Charge_Ratio (derived)	Billed divided by Paid. A 5x ratio = hospital billed 5x what Medicare paid.

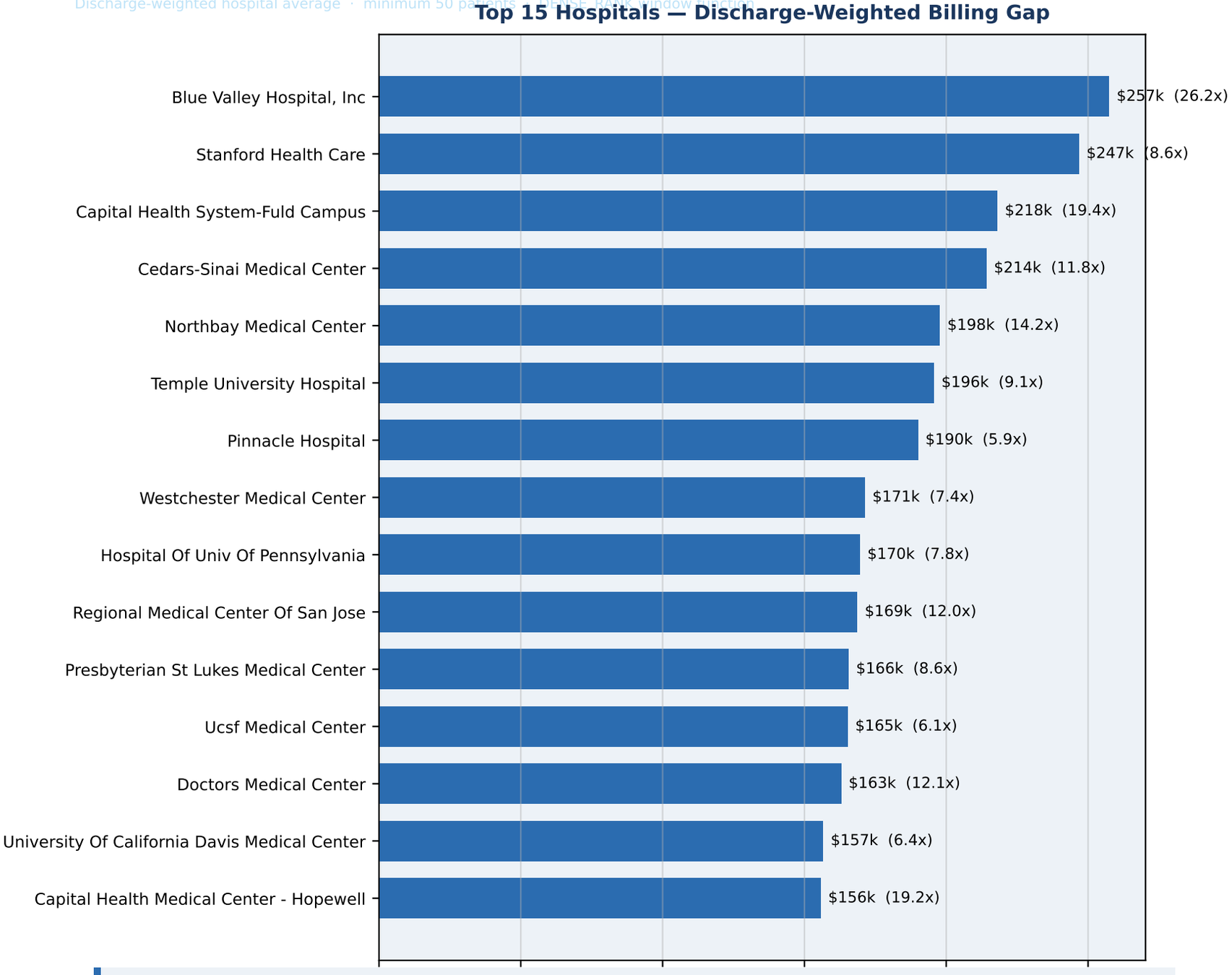
T2 DRG Details — the procedure glossary

DRGV22	DRG code — joins to DRG_Cd in T1.
MDC / MDC_Name	Major Diagnostic Category: 25 body-system buckets (MDC 05 = Heart/Circulatory).
DRG Title	Human-readable name of the procedure type.
Relative Weights	CMS complexity score. Higher = more resource-intensive = higher Medicare payment.
Geometric Mean LOS	Typical hospital stay in days.

T3 Provider Info — the hospital profile

Facility ID	6-digit hospital ID — joins to CCN in T1.
Hospital Type	Acute Care · Critical Access (rural 25 beds) · Psychiatric · Childrens.
Hospital Ownership	Voluntary non-profit · Proprietary (for-profit) · Government.

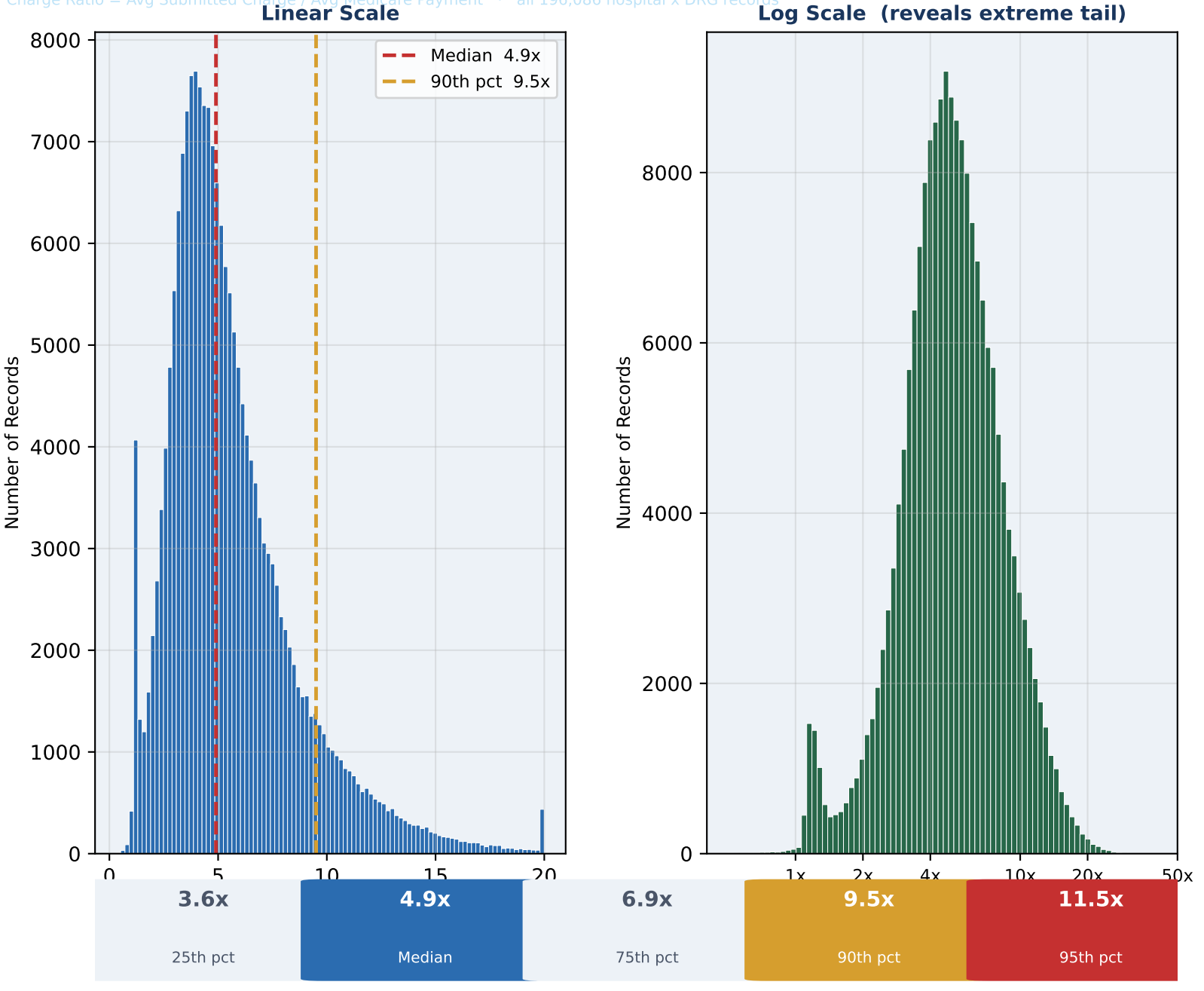
Discharge-weighted hospital average · minimum 50 patients · DENSE_RANK window function



What This Tells Us

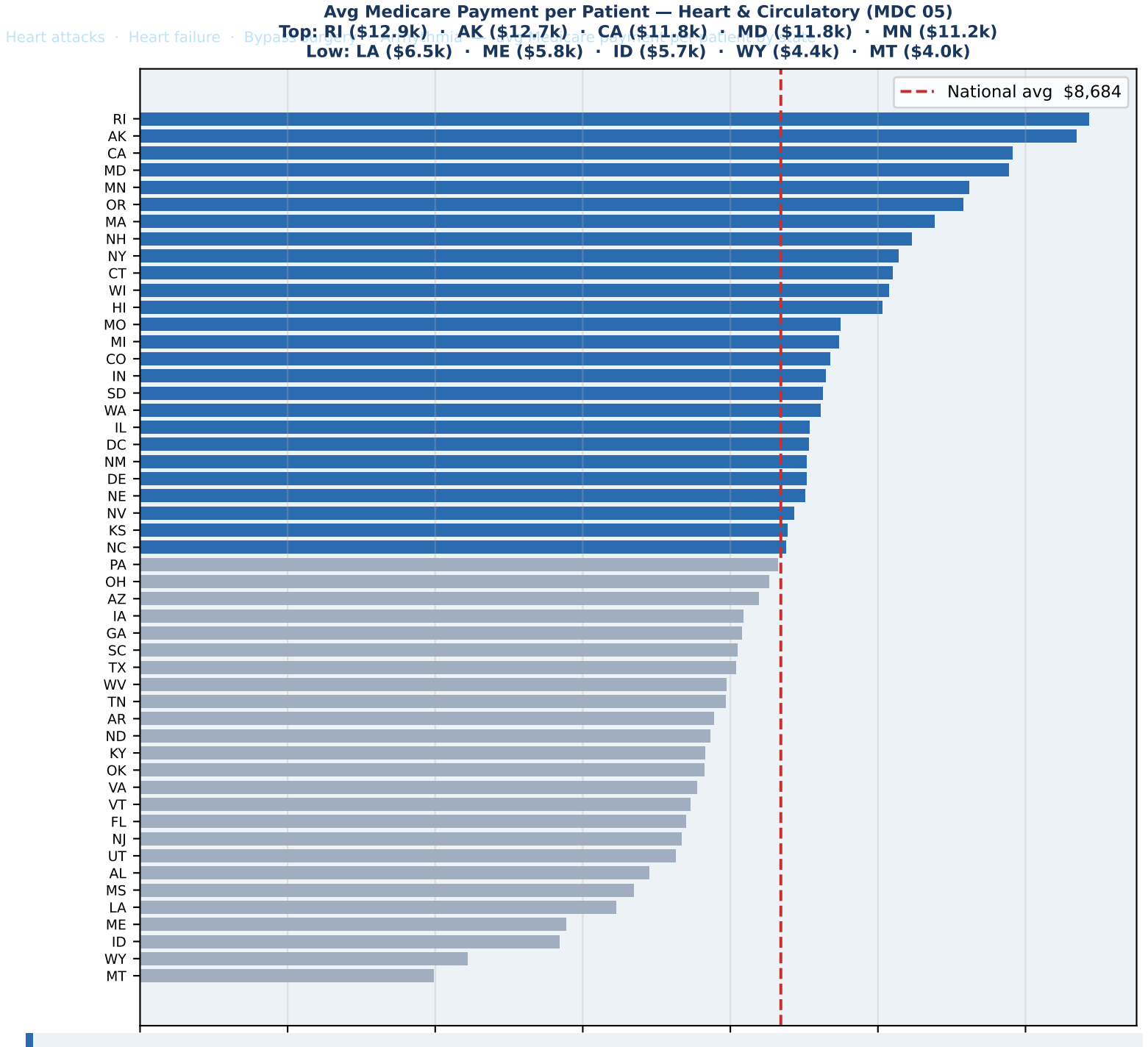
Billing Gap = Avg Submitted Charge minus Avg Medicare Payment — the dollar amount billed above what Medicare paid per patient. A discharge-weighted average (SUM(gap x patients) / SUM(patients)) is used so high-volume procedures carry more weight. Stanford, Cedars-Sinai, and several California/New Jersey hospitals lead — reflecting complex case mix at academic centres, high-cost geography (wage index), and aggressive charge-master pricing. A $250k billing gap does not mean Medicare overpaid — it means the hospital sticker price is $250k above what it actually received.

Charge Ratio = Avg Submitted Charge / Avg Medicare Payment · all 196,086 hospital x DRG records



What This Tells Us

The charge ratio is a scale-free multiplier — it lets you compare a $2,000 procedure and a $200,000 one on equal footing.
LEFT (linear): Most records cluster between 3x and 7x. The typical US hospital bills 4-5 times what Medicare pays.
RIGHT (log scale): Reveals a long right tail the linear chart hides. A small fraction reach 20x, 30x, even 50x.
These extreme outliers are often small hospitals with low Medicare volumes using aggressive charge-master pricing.
The distribution is strongly right-skewed — the mean is pulled far above the median by outliers, which is why percentiles give a more honest picture of the typical hospital than a simple average.

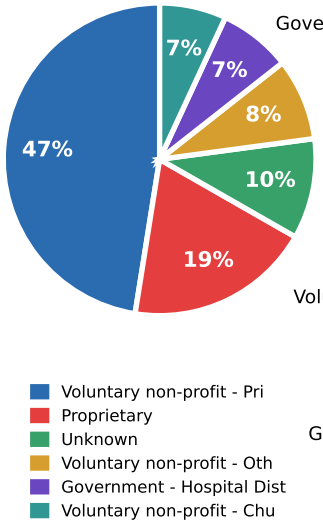


What This Tells Us

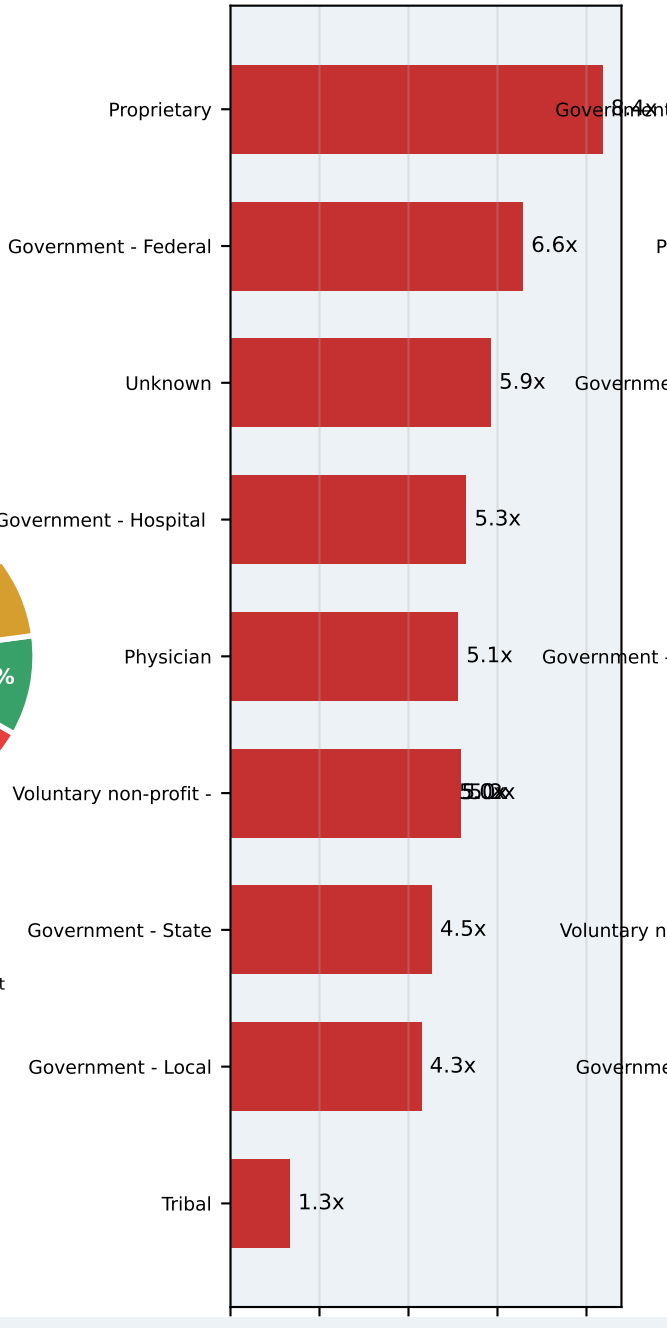
MDC 05 (Circulatory — heart attacks, bypass surgery, arrhythmia) is the #1 body system by total Medicare spend. States like Rhode Island, California, and Maryland pay 30-40% MORE per heart patient than the national average. This is not over-billing. It reflects: (1) Wage Index — Medicare adjusts payments where labour costs are high; (2) Academic medical centres receive higher base rates; (3) DSH bonus payments for hospitals serving low-income patients. High-volume states (CA, TX, FL) dominate total spend simply due to population size — that is a different story from per-patient cost. Both views together give the full picture.

How does ownership type correlate with charge ratio and billing gap?

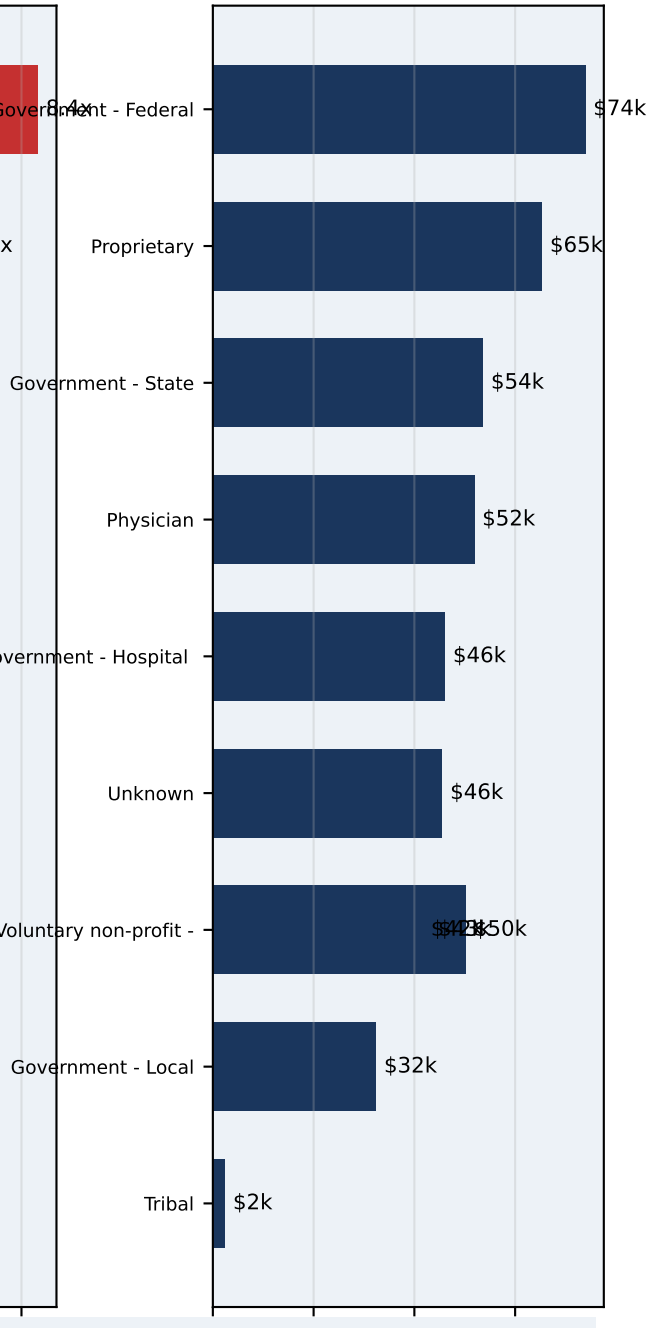
Hospital Count by Ownership



Charge Ratio by Ownership



Billing Gap by Ownership

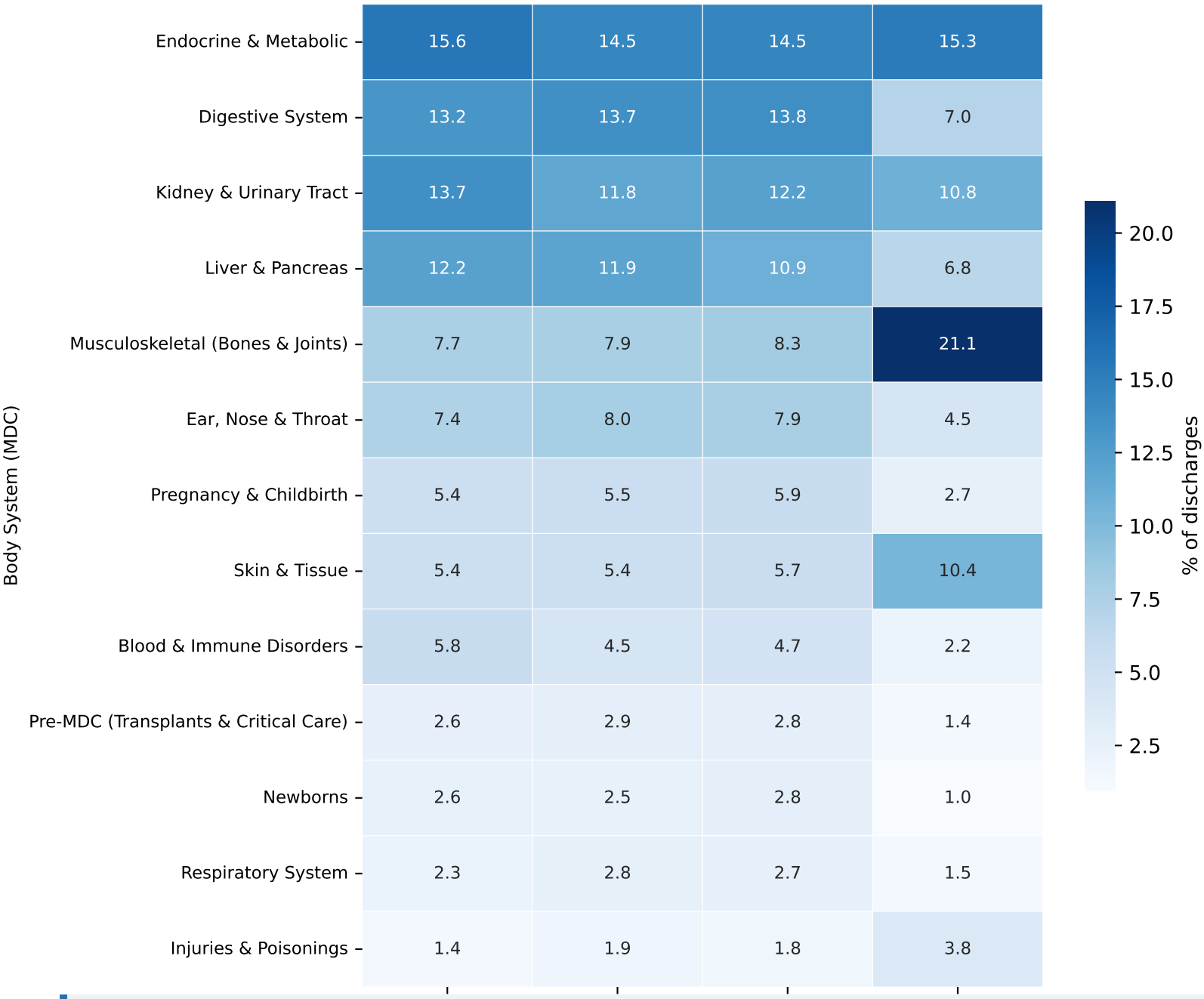


What This Tells Us

Voluntary non-profit private hospitals are the most common type (~42% of hospitals). They reinvest all income back into care rather than distributing profits — and their charge ratios are consistently lower than for-profit.
For-profit (Proprietary) hospitals consistently show the HIGHEST charge ratios and billing gaps.
This is a structural difference tied to shareholder return obligations, not differences in patient type.
Government hospitals (VA, state, local) show the LOWEST charge ratios — no profit motive, often safety-net.
The gap between for-profit and non-profit is measurable and consistent across all DRG types.

What % of each ownership type's patients fall into each body system category?

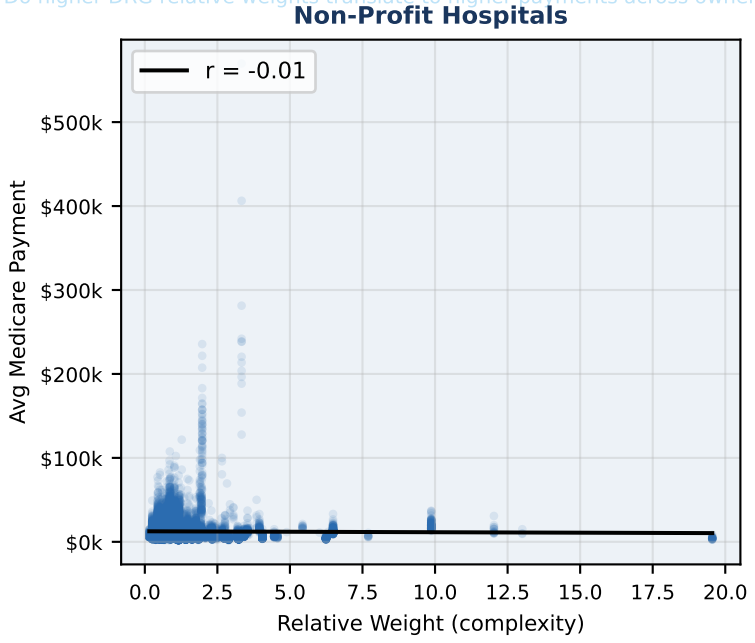
Discharge Mix (%) by Hospital Ownership — cell = % of that ownership type's total patients

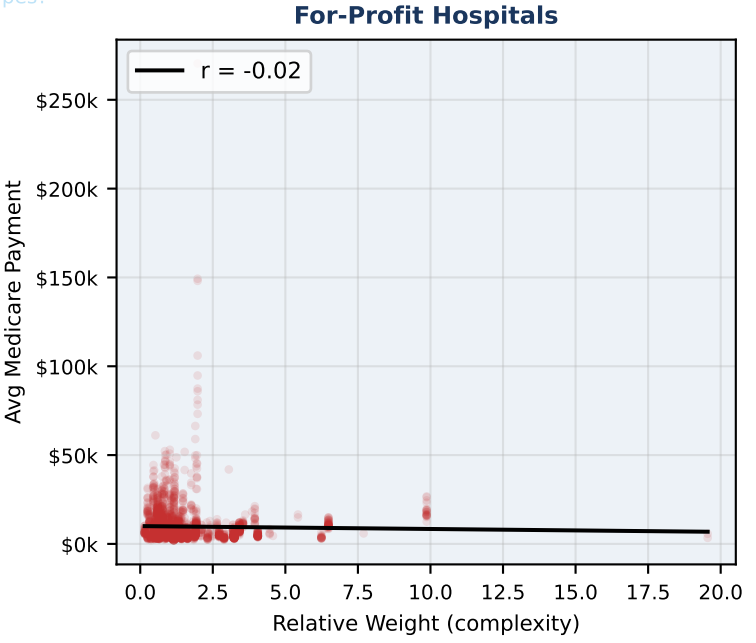


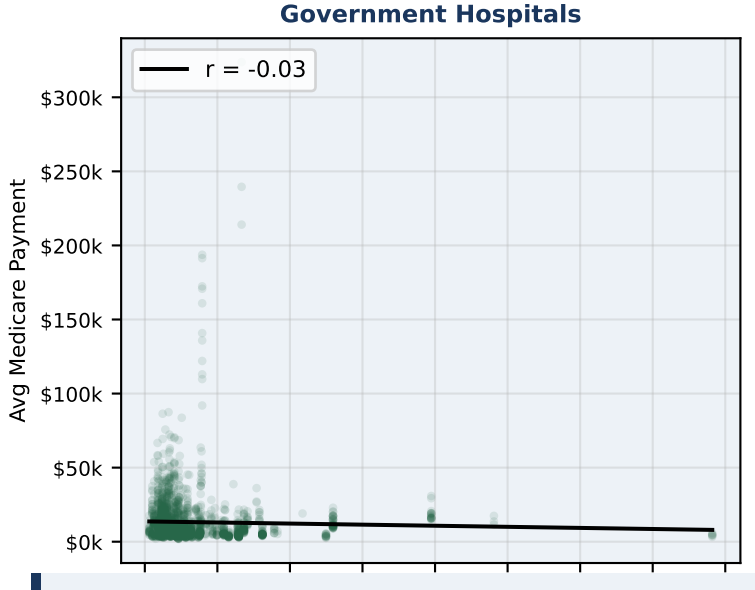
What This Tells Us

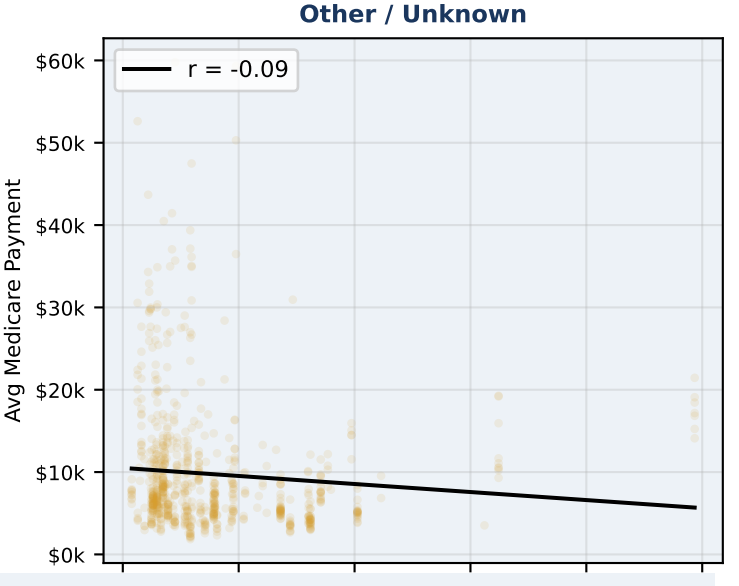
All three ownership types are primarily Acute Care hospitals, so the MDC mix is broadly similar — but differences exist at the margins. For-profit hospitals have a slightly higher share of Musculoskeletal (elective joint replacements) and Digestive procedures relative to Government hospitals.
Government hospitals show a higher proportion of Nervous System and Circulatory cases — consistent with their role as safety-net providers treating more complex and uninsured patients.
Non-profit hospitals are the most balanced across all body systems, reflecting their broader community mission.

Do higher DRG relative weights translate to higher payments across ownership types?





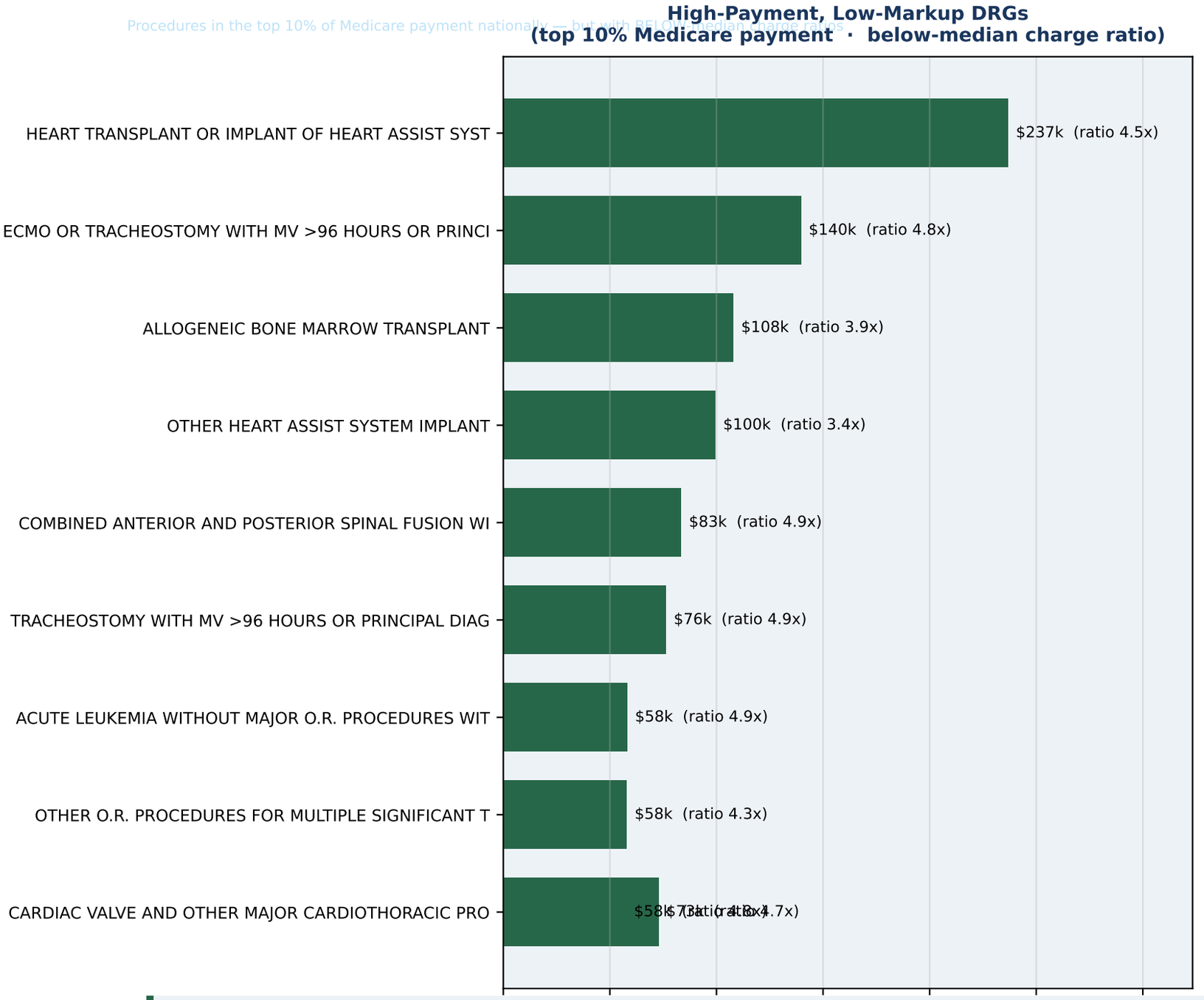




What This Tells Us

DRG relative weight vs Medicare payment correlations: Non-Profit r=-0.01, For-Profit r=-0.02, Government r=-0.03. The weak correlations in this hospital-level scatter reflect the high within-DRG payment variance across hospitals: the same procedure code pays very differently depending on wage index, DSH adjustments, and outlier payments. For-profit hospitals tend to have a higher payment intercept for identical DRG complexity — consistent with their more aggressive charge-master pricing and higher base charges at each complexity level. The complexity-payment relationship is best seen at the national DRG-average level, not hospital-by-hospital.

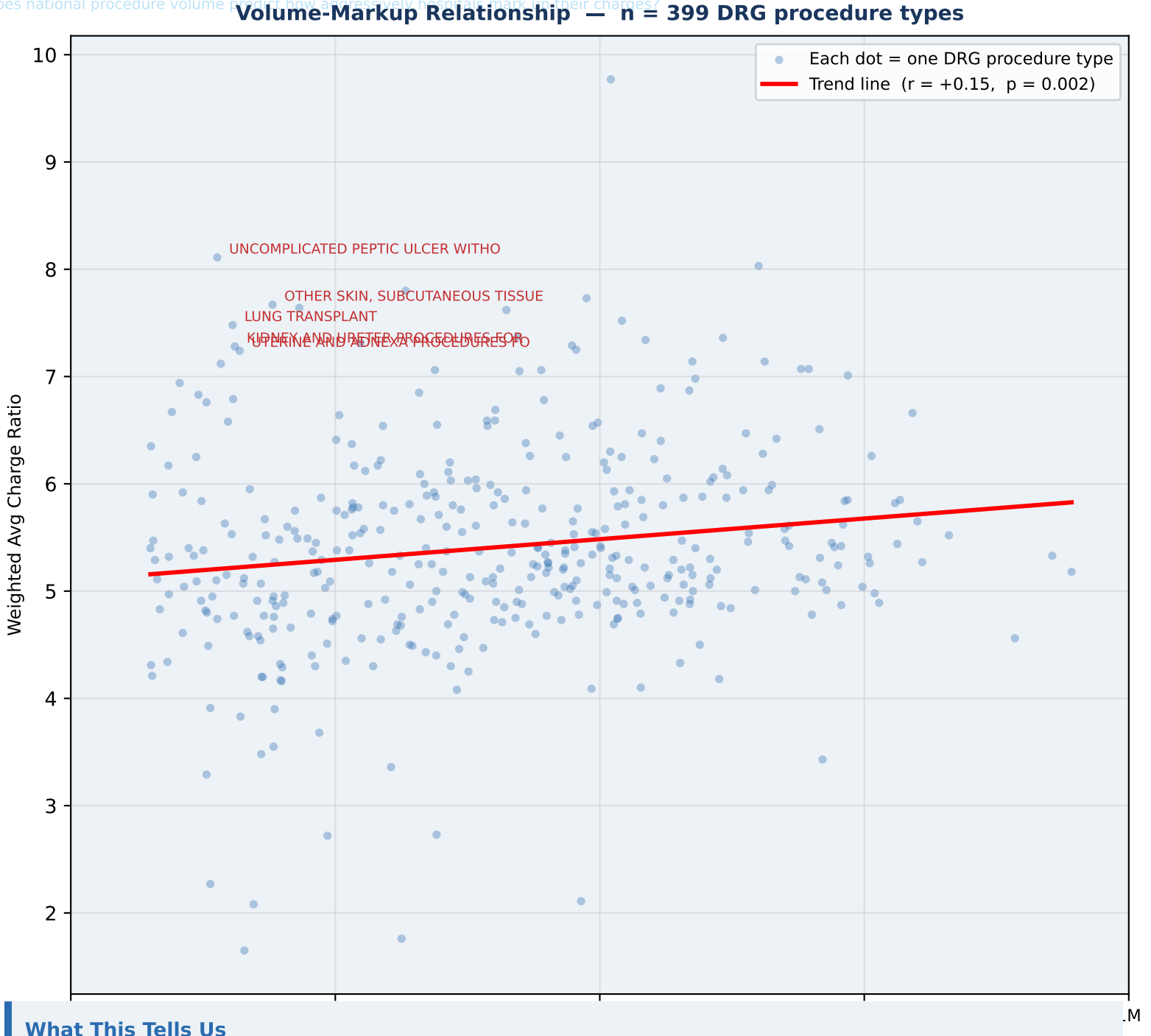
Procedures in the top 10% of Medicare payment nationally — but with BELOW-median charge ratios



What This Tells Us

Most billing-gap analysis focuses on hospitals charging extreme multiples. This finding asks the opposite: which procedures does Medicare pay the most for, even though hospitals are NOT marking them up aggressively? Heart transplants (~$237k), ECMO — mechanical life support (~$140k), and bone marrow transplants (~$108k) are in the top 10% of all Medicare payments nationally, yet their charge ratios are BELOW the median. This is not overbilling — it is policy-driven reimbursement. CMS deliberately pays generously to ensure hospitals can afford to keep performing these life-saving procedures. Flagging these purely for high billing gaps is misleading.

Does national procedure volume predict how aggressively hospitals mark up their charges?

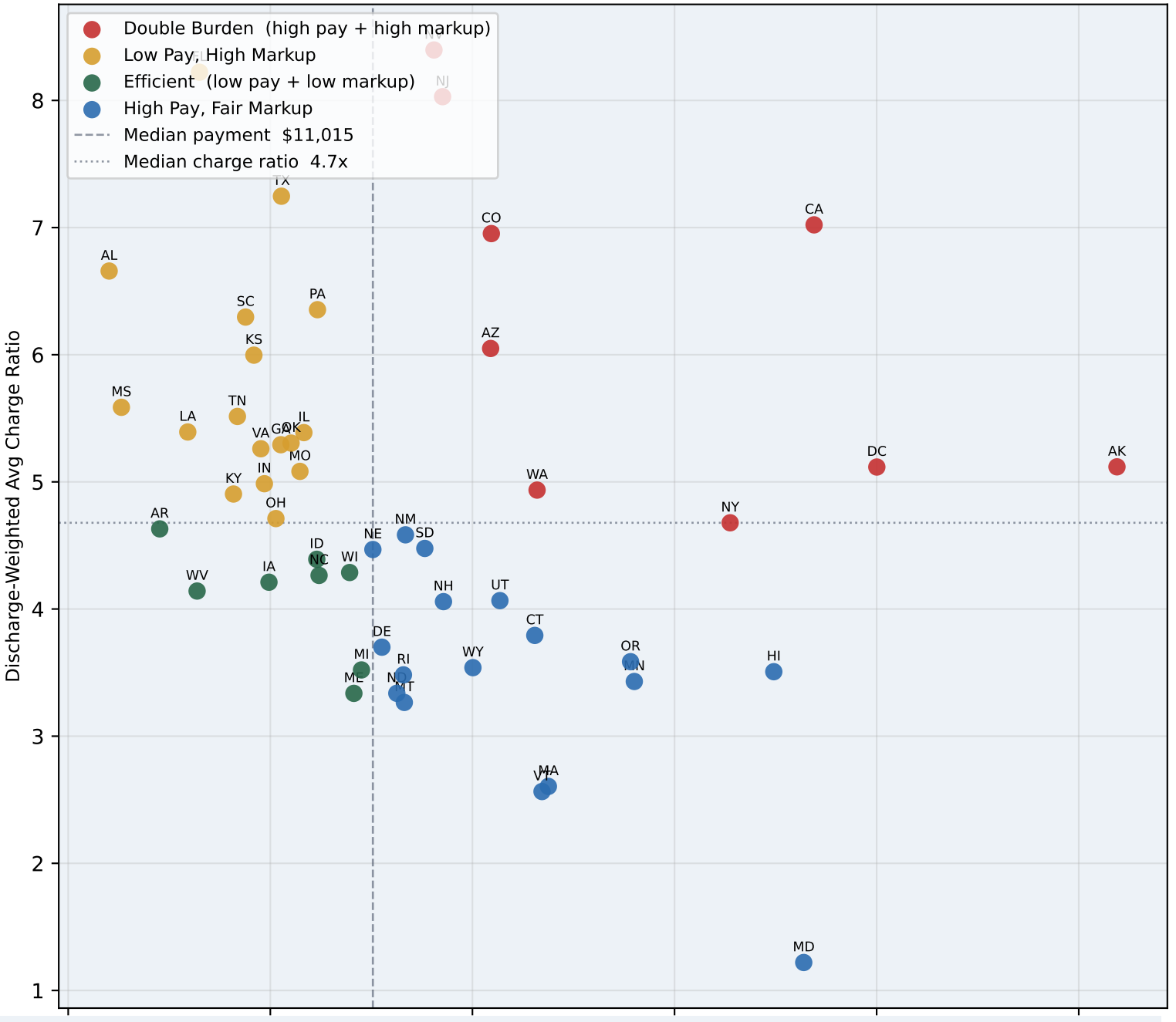


What This Tells Us

The correlation between national procedure volume and charge ratio is r = +0.15 — a weak but statistically significant POSITIVE relationship. This is driven by the extreme right tail: rare procedures performed at only a handful of centres (visible as annotated outliers with very low volume) show charge ratios of 15x-40x. Common procedures (hip/knee replacements, simple pneumonia, fractures) cluster at 3-5x. Although the aggregate correlation is weak-positive, the outlier pattern is clear: the most extreme markups occur in rare, hard-to-compare procedures — a market structure effect, not clinical necessity.

Which states have BOTH above-median Medicare payment AND above-median charge ratio?

State Payment-Markup Matrix — upper-right = both high payment AND high markup



What This Tells Us

This scatter divides all US states into four quadrants: (1) Double Burden — high payment AND high markup; (2) High Pay, Fair Markup — expensive because of wage index and academic centres, not billing aggression; (3) Low Pay, High Markup — hospitals compensating for lower base rates with aggressive charge-master pricing; (4) Efficient — below-median on both dimensions.

States like California and New York land in "High Pay, Fair Markup" — Medicare pays more because labour costs are genuinely higher. The "Double Burden" states are the clearest signal for targeted regulatory attention: these are markets where both Medicare overpays AND hospitals still markup aggressively.

Task 01 | Data Understanding

Three CMS datasets · 196,086 hospital x DRG records · 3,182 hospitals · 51 states/territories. The row grain is hospital x procedure— critical for correct discharge-weighted averages. Billing_Gap and Charge_Ratio were derived, not provided in the raw data.

Task 02 | The Billing Gap

Median charge ratio 4.9x — hospitals bill 4.9 times what Medicare pays. Top 5% exceed 11.5x. Geographic variation in MDC 05 (heart): $12.9k (RI) vs $4.0k (MT) — driven by wage index and academic centre concentration, not billing aggression.

Task 03 | Ownership & Hospital Type

For-profit hospitals consistently show higher charge ratios than non-profits and government hospitals. MDC mix analysis proves hospital types treat fundamentally different patient populations — raw billing comparisons require case-mix adjustment. DRG complexity strongly predicts payment across all ownership types, validating that the DRG weighting system works as intended.

Task 04 | Three Independent Findings

(A) Policy-Driven Payments: heart transplants & ECMO pay $100k-$250k with below-median charge ratios— CMS pays generously by design. (B) Volume-Markup: rare procedures show the most extreme charge ratios (15x-40x) vs. common procedures at 3-5x— a market structure effect, not clinical necessity. (C) State Double-Burden: some states have both above-median payments AND markups— the clearest signal for targeted regulatory attention.